JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Mathematical Induction

JEE Main Mathematics · Previous Year Questions with Answer Key

3	3	2002 – 2005	3	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Out of syllabus	3 questions
Answer key	at the end

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.

Out of syllabus 3 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q1. AIEEE 2005 · MCQ (Single Correct Answer)

If $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then which one of the following holds for all $n \geq 1$, by the principle of mathematical induction?

(A) $A^n = nA - (n-1)\mathrm{I}$

(B) $A^n = 2^{n-1}A - (n-1)\mathrm{I}$

(C) $A^n = nA + (n-1)\mathrm{I}$

(D) $A^n = 2^{n-1}A + (n-1)\mathrm{I}$

Q2. AIEEE 2004 · MCQ (Single Correct Answer)

Let $S(K) = 1 + 3 + 5 \ldots + (2K-1) = 3 + K^2$. Then which of the following is true

(A) Principle of mathematical induction can be used to prove the formula

(B) $S(K) \Rightarrow S(K+1)$

(C) $S(K) \neq S(K+1)$

(D) $S(1)$ is correct

Q3. AIEEE 2002 · MCQ (Single Correct Answer)

If $a_n = \sqrt{7 + \sqrt{7 + \sqrt{7 + \ldots\ldots}}}$ having n radical signs then by methods of mathematical induction which is true

(A) $a_n > 7 \ \forall \ n \geq 1$

(B) $a_n < 7 \ \forall \ n \geq 1$

(C) $a_n < 4 \ \forall \ n \geq 1$

(D) $a_n > 3 \ \forall \ n \geq 1$

Answer Key

1. **A**	2. **B**	3. **D**